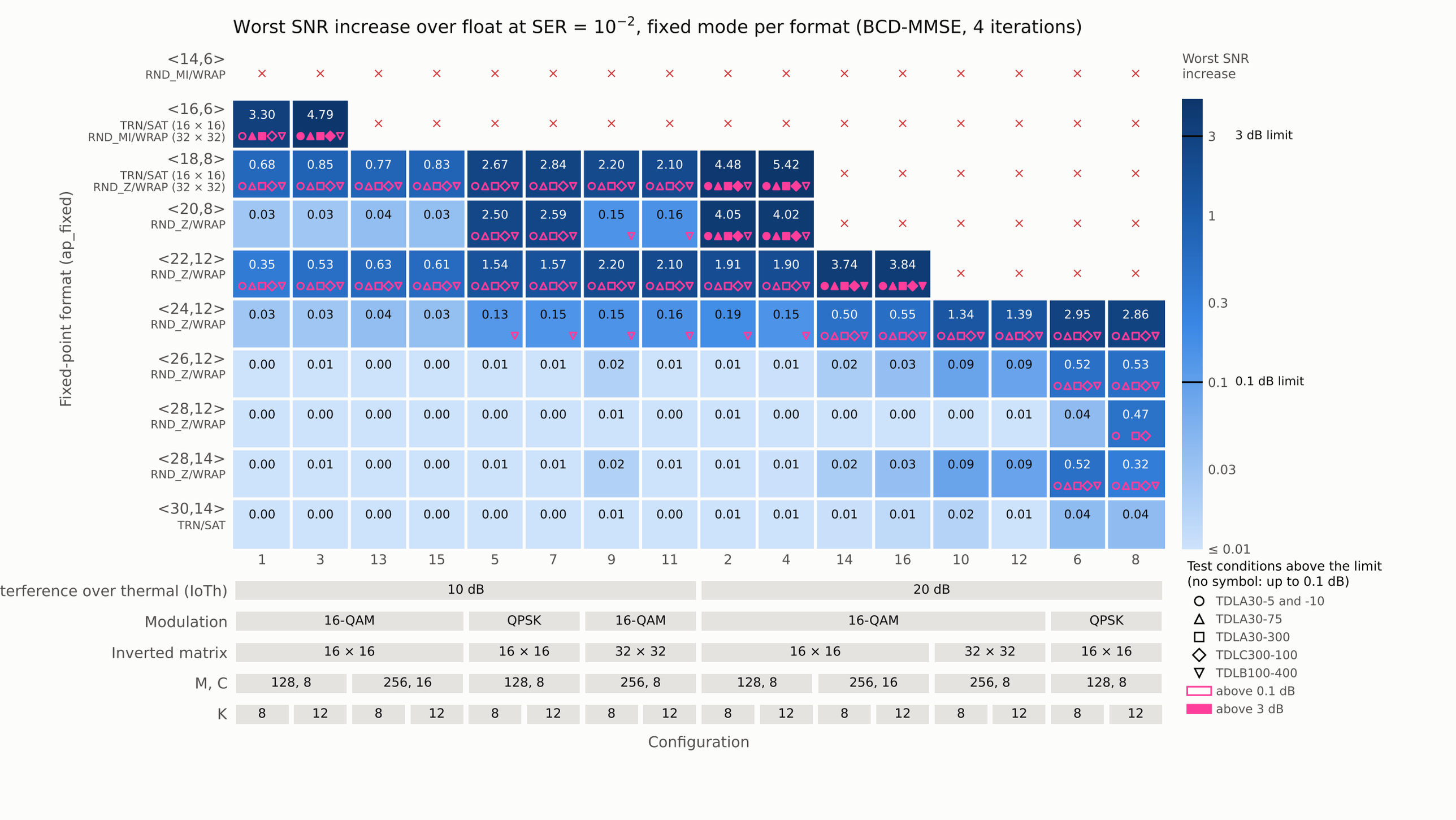
Worst SNR increase over float at SER = 10^{-2}, fixed mode per format (BCD-MMSE, 4 iterations)
Fixed-point format (ap_fixed)
<14,6> RND_MI/WRAP
<16,6> TRN/SAT (16 × 16) RND_MI/WRAP (32 × 32)
<18,8> TRN/SAT (16 × 16) RND_Z/WRAP (32 × 32)
<20,8> RND_Z/WRAP
<22,12> RND_Z/WRAP
<24,12> RND_Z/WRAP
<26,12> RND_Z/WRAP
<28,12> RND_Z/WRAP
<28,14> RND_Z/WRAP
<30,14> TRN/SAT
3.30 4.79
0.68 0.85 0.77 0.83 2.67 2.84 2.20 2.10 4.48 5.42
0.03 0.03 0.04 0.03 2.50 2.59 0.15 0.16 4.05 4.02
0.35 0.53 0.63 0.61 1.54 1.57 2.20 2.10 1.91 1.90 3.74 3.84
0.03 0.03 0.04 0.03 0.13 0.15 0.15 0.16 0.19 0.15 0.50 0.55 1.34 1.39 2.95 2.86
0.00 0.01 0.00 0.00 0.01 0.01 0.02 0.01 0.01 0.01 0.02 0.03 0.09 0.09 0.52 0.53
0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.00 0.01 0.00 0.00 0.00 0.00 0.01 0.04 0.47
0.00 0.01 0.00 0.00 0.01 0.01 0.02 0.01 0.01 0.01 0.02 0.03 0.09 0.09 0.52 0.32
0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.00 0.01 0.01 0.01 0.01 0.02 0.01 0.04 0.04
1 3 13 15 5 7 9 11 2 4 14 16 10 12 6 8
Worst SNR increase
3
3 dB limit
1
0.3
0.1
0.1 dB limit
0.03
≤ 0.01
Test conditions above the limit (no symbol: up to 0.1 dB)
TDLA30-5 and -10
TDLA30-75
TDLA30-300
TDLC300-100
TDLB100-400
above 0.1 dB
above 3 dB
terference over thermal (IoTh)
10 dB
20 dB
Modulation
16-QAM
QPSK
16-QAM
16-QAM
QPSK
Inverted matrix
16 × 16
16 × 16
32 × 32
16 × 16
32 × 32
16 × 16
M, C
128, 8
256, 16
128, 8
256, 8
128, 8
256, 16
256, 8
128, 8
K
8 12 8 12 8 12 8 12 8 12 8 12 8 12 8 12
Configuration